

DCS — Pinout & Terminal Connections

Exact wiring: Pi GPIO (via OONO TB) · ADS1115 · shunt · Rosemount loop

RASPBERRY PI 40-PIN (BCM) via OONO GPIO TB

Pi pin 1 (3V3) → ADS1115 VDD
Pi pin 3 (GPIO2/SDA) → ADS1115 SDA
Pi pin 5 (GPIO3/SCL) → ADS1115 SCL
Pi pin 6 (GND) → ADS1115 GND · shunt low
Pi pin 11 (GPIO17) → Relay IN (pump, optional)
ADDR: ADS1115 ADDR → GND ⇒ 0x48
I2C bus: 1 (i2cdetect -y 1)

ADS1115 BREAKOUT

VDD ← 3V3
GND ← Pi GND / shunt low
SCL ← GPIO3
SDA ← GPIO2
ADDR ← GND (0x48)
AIN0 ← shunt high (sense +)
AIN1..3 unused (tie to GND optional)
PGA: ±4.096 V (software default)

4–20 mA LOOP / SHUNT / PROTECTION

+24 VDC (HDR-15-24) → [optional PT 4-HESI fuse TB] → Rosemount +
Rosemount – → shunt high → ADS1115 AIN0
shunt low → 24 VDC – AND ADS1115 GND / Pi GND (single star)
 $R_s = 100\ \Omega\ 0.1\% \Rightarrow 4\ \text{mA} \rightarrow 0.40\ \text{V}\quad 20\ \text{mA} \rightarrow 2.00\ \text{V}\quad | \quad \text{Scale: LRV@4 mA, URV@20 mA (default 0–100 °C)}$

ASSEMBLY OPTIONS

- A. Breadboard – Pi + OONO GPIO TB, ADS1115 module, shunt across AIN0/GND, leads to transmitter / 24 V.
- B. Junction box / soldered – same netlist; fuse TB on loop, DIN PSU, screw terminals for field cable.
- C. 5 V ADS only – power ADS from 5 V and pass SDA/SCL through Adafruit 757 level shifter to Pi 3V3 I2C.
